Git Guide: Brad Sherlock

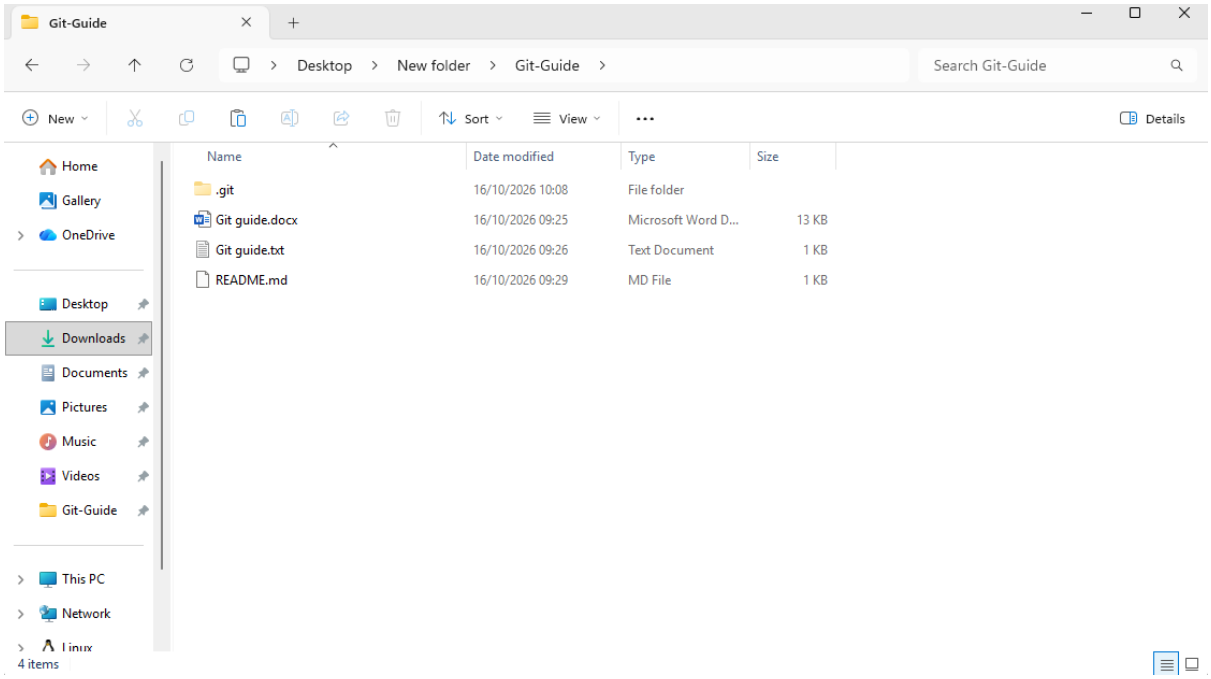
Git init

[-b <*branch-name*> | --initial-branch=<*branch-name*>]

Git init creates an empty git repository or reinitializes an existing one. Branch name is where the name of the branch is inputted. If no name is given name will default to master/main.

```
C00336705@CD-A307-18 MINGW64 /c/Users/C00336705/Desktop/New folder/Git-Guide (main)
$ git init
Reinitialized existing Git repository in C:/Users/C00336705/Desktop/New folder/Git-Guide/.git/

C00336705@CD-A307-18 MINGW64 /c/Users/C00336705/Desktop/New folder/Git-Guide (main)
$
```



Git add

git add “File Name” (Exactly as saved)

Adds content from new or changed files to the index (or staging area). This is used to prepare for next commit.

<u>Wrong</u>

```
C00336705@CD-A307-18 MINGW64 /c/Users/C00336705/Desktop/New folder/Git-Guide (main)
$ "git add Git guide.txt"
bash: git add Git guide.txt: command not found
```

<u>Right</u>

```
C00336705@CD-A307-18 MINGW64 /c/Users/C00336705/Desktop/New folder/Git-Guide (main)
$ git add "Git guide.txt"
```

Git status

Shows the working tree/working directory status. E.g. added files or modified files in working area.

```
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ~$t guide.docx

nothing added to commit but untracked files present (use "git add" to track)
```

Git log

git log [<*options*>]

Lists all commits reachable by following the parent links from the given commit(s). Excludes commits reachable from the one(s) given with a ^ in front of them. Lists the items from most recent to oldest.

```
c00336705@CD-A002-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git log
commit 6118abc22ec2b92371914b61e8d34adb782183b4 (HEAD -> main, origin/main, origin/HEAD)
Author: (Student  C00336705) Brad Sherlock <C00336705@setu.ie>
Date:   Fri Oct 16 15:34:01 2026 +0100

    a

commit f281b997ded06b3a5171d047506c88d3db9f1e3f
Author: (Student  C00336705) Brad Sherlock <C00336705@setu.ie>
Date:   Fri Oct 16 10:42:51 2026 +0100

    all

commit 2bf78abe72529e161381e33dfc3aa19c06075426
Author: (Student  C00336705) Brad Sherlock <C00336705@setu.ie>
Date:   Fri Oct 16 10:38:43 2026 +0100

    Git guide.txt

commit 29ae5db1998142587973c15412fc2de0f7031ace
Author: (Student  C00336705) Brad Sherlock <C00336705@setu.ie>
Date:   Fri Oct 16 10:24:18 2026 +0100

    Git guide.txt

commit b1b7928c651dad34724a6666d39cf7d09f8b11a5
Author: (Student  C00336705) Brad Sherlock <C00336705@setu.ie>
Date:   Fri Oct 16 09:31:58 2026 +0100

    first commit
```

Git commit

git commit [-a | (all) / git commit -m “name” / git commit -am “name” (all and name)

Creates new commit containing current contents of the index and the given log message showing the changes. New commit is a direct child of HEAD, usually the tip of current branch, and the branch is updated to point to it.

```
c00336705@CD-A002-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git commit -am "Git log"
[main 53e5923] Git log
 Committer: Brad-Sherlock <C00336705@setu.ie>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 3 files changed, 18 insertions(+), 1 deletion(-)
 create mode 100644 ~WRL1192.tmp
```

Git push

Git push -u origin main (first) / git push (all after first)

Updates one or more branches, nodes, etc in remote repository with information from local repository. Sends all necessary data that isn’t already on remote. Must use git push origin -u main for first push.

```
c00336705@CD-A002-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 20 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 25.66 KiB | 12.83 MiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Brad-Sherlock/Git-Guide.git
   53e5923..28fb8b8  main -> main
```

Git config

git config user.name “Name”/git config user.email “email”/git config--global core.editor notepad

git config is used to get and set global options or repositories. Can be used to set default name and email for commits and can be used to set windows notepad as default editor.

```
c00336705@CD-A002-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git config --global user.name "Brad-Sherlock"
```

Git remote

Git remote –v/Git remote add <name> <url>/Git remote rm <name>

Manages set of repositories who's branches you track. Can also be used to list, create and destroy remote repositories.

```
c00336705@CD-A003-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git remote
origin
```

Git branch

Git branch new_branch_name / Git branch –a/-l/-r / Git ls-remote

Git branch lists, creates or delete branches. This includes creating new branches in local repositories, listing all branches in repositories and listing branches on remotes (including unknowns).

```
c00336705@CD-A003-09 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git branch
* main
```

Git merge

Git merge joins 2 or more development histories together by incorporating changes from the named commits into the current branch.

```
c00336705@CD-A002-13 MINGW64 /c/Users/c00336705/Desktop/GitGuide/Git-Guide (main)
$ git merge branch1
Updating aa189b1..5b96673
Fast-forward
 dontReadPete.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 dontReadPete.txt
```